

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0701

Gallaway 138 kV
April 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0701. The Interconnection Customer proposes a 114.3 MW solar photovoltaic facility interconnecting at the Gallaway 138 kV substation in the Ohio area. The request was received on November 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 284 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$159,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STEWART MTN DAM to RITTER PARK 345kV Line
Contingency: P.T.S to MEDORA 345kV
Loading: 101.7% (Rating: 1,471 MVA)
DFAX: 8.9%
Pre-project loading: 101.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STEWART MTN DAM to RITTER PARK 345kV	Reconductor line	\$8,168,000	2% (\$159,000)
	Total allocated to project		\$159,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.